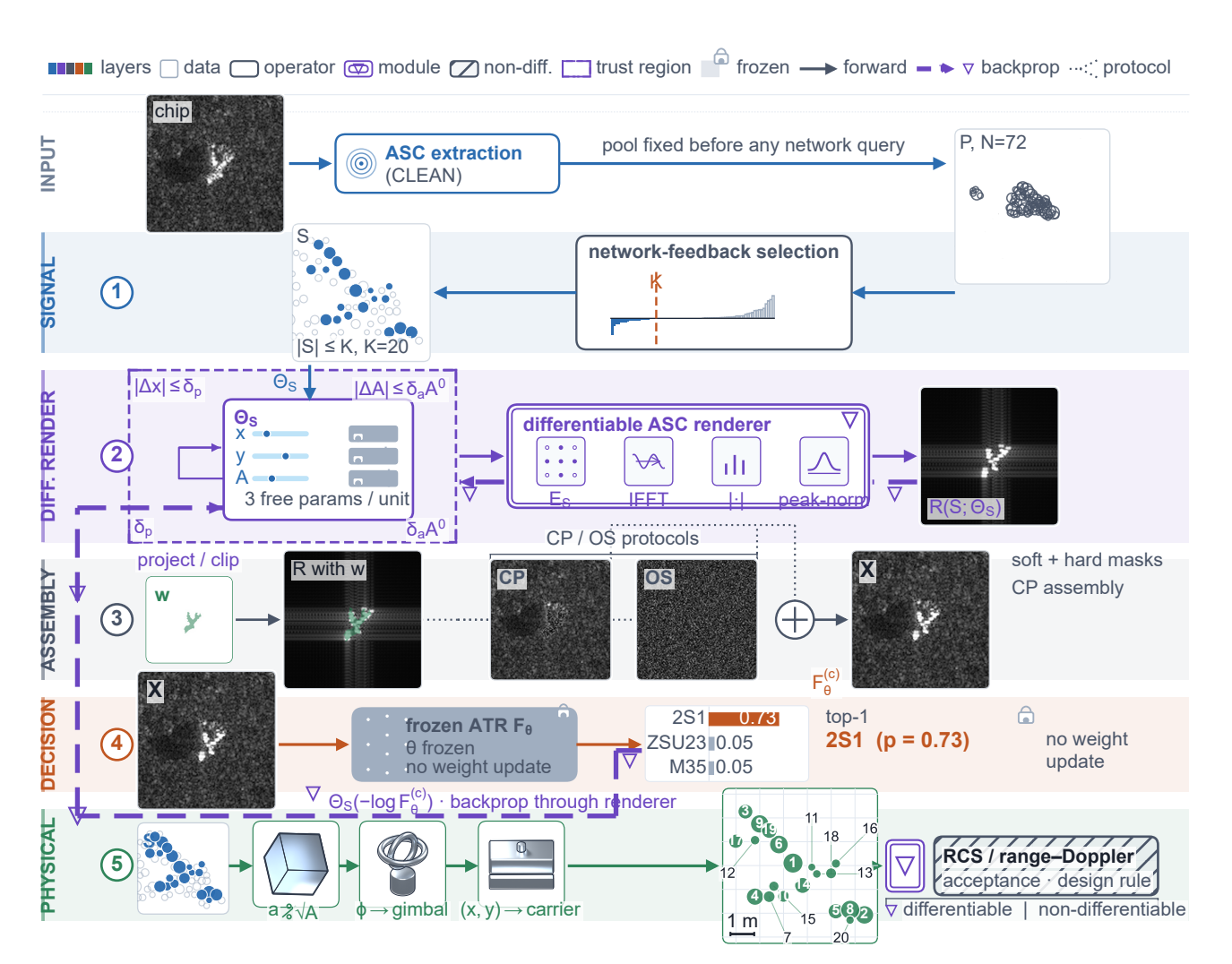

layers
data
operator
module
non-diff.
trust region
frozen
forward
backprop
protocol
INPUT
chip
ASC extraction
(CLEAN)
pool fixed before any network query
P, N=72
SIGNAL
1
S
|S| ≤ K, K=20
network-feedback selection
K
DIFF. RENDER
2
|Δx| ≤ δp
Θs
|ΔA| ≤ δaA0
Θs
x
y
A
3 free params / unit
δp
δaA0
differentiable ASC renderer
Es
IFFT
|·|
peak-norm
R(S; Θs)
ASSEMBLY
3
project / clip
w
R with w
CP / OS protocols
CP
OS
X
soft + hard masks
CP assembly
DECISION
4
X
frozen ATR Fθ
θ frozen
no weight update
2S1 0.73
ZSU23 0.05
M35 0.05
top-1
2S1 (p = 0.73)
no weight
update
Θs(−log F(c)θ) · backprop through renderer
PHYSICAL
5
S
a ∝ √A
ϕ → gimbal
(x, y) → carrier
1 m
RCS / range–Doppler
acceptance · design rule
differentiable | non-differentiable